Project description:
Skills and Personality (SKIPPER)

The Authors

DISCLAIMER

The following registration refers to the general study design/procedure and research questions of the research project “Skills and Personality”. Specific study projects, hypotheses and planned statistical analyses will be registered separately elsewhere.

BACKGROUND

Previous research has shown that individuals have a desire to change aspects of their personality (Baranski et al., 2021; Hudson & Roberts, 2014; Thielmann & Vries, 2021) and can do so to a small extent (Hudson, 2021; Hudson & Fraley, 2015). *Personality traits* are relatively stable patterns of behaviors, feelings and thoughts (Funder, 2001; Roberts & Yoon, 2022; Soto, 2018) and are said to be changeable through modifications of existing preferences and habit formation (Soto et al., 2022b). On the other hand, it is assumed that changes in s*ocial, emotional, and, behavioral skills* (SEB) are possible by *learning* (new) behaviors (Soto et al., 2022b). By learning we mean more conscious meta-cognitive learning in addition to exhibiting new behavior. SEB skills are *conceptually different* from personality traits. They are defined as functional capacities for behaviors in critical situations; that is, what a person is capable of doing when a situation calls for it (Lechner et al., 2022; Paulhus & Martin, 1987; Soto

et al., 2021). Overall, five broad skill domains are distinguished: self-management, emotional resilience, social engagement, cooperation, and innovation (Soto et al., 2021; Soto et al., 2022b). To our knowledge, there is no empirical information on the malleability of SEB skills in adulthood. The main aim of this research project is to investigate whether individuals can improve skills by learning, given their willingness to do so (Research question 1, RQ1). Another key objective is to explore whether psychological motives facilitate this effect, as they are supposed to select and energize relevant behaviors (RQ2) (McClelland, 1985). Additionally, we are interested in whether SEB skills exhibit greater malleability than personality traits (RQ3).

We focus on the following two of the SEB skills: social engagement and emotional resilience. *Social engagement skills (SE)* refer to the capacity to actively engage with others, while *emotional resilience skills (ER)* encompass capacities to regulate emotions (Soto et al., 2021). These SEB skills were selected due to their empirical overlap with the personality traits Extraversion and Emotional Stability (Lechner et al., 2022; Soto et al., 2022b), for which the majority of individuals express a desire to change (Baranski et al., 2021; Hudson & Roberts, 2014; Thielmann & Vries, 2021).

Two learning interventions, one for each skill, were developed following a content analyses of both skills (a detailed description of the development process is not provided here). The process of the intervention development relies on proposed principles for smart teaching that were derived from learning principles (Ambrose et al., 2010) and instructional design (Gagné, 1985; van Merriënboer & Kirschner, 2018). The structure of the intervention follows Gagnés (1985) nine steps of learning:

1. Gain attention

2. Information on learning objectives
3. Activation of prior knowledge
4. Present the learning stimulus
5. Provide a learning guidance
6. Practice
7. Provide feedback
8. Evaluate performance
9. Enhance retention and transfer

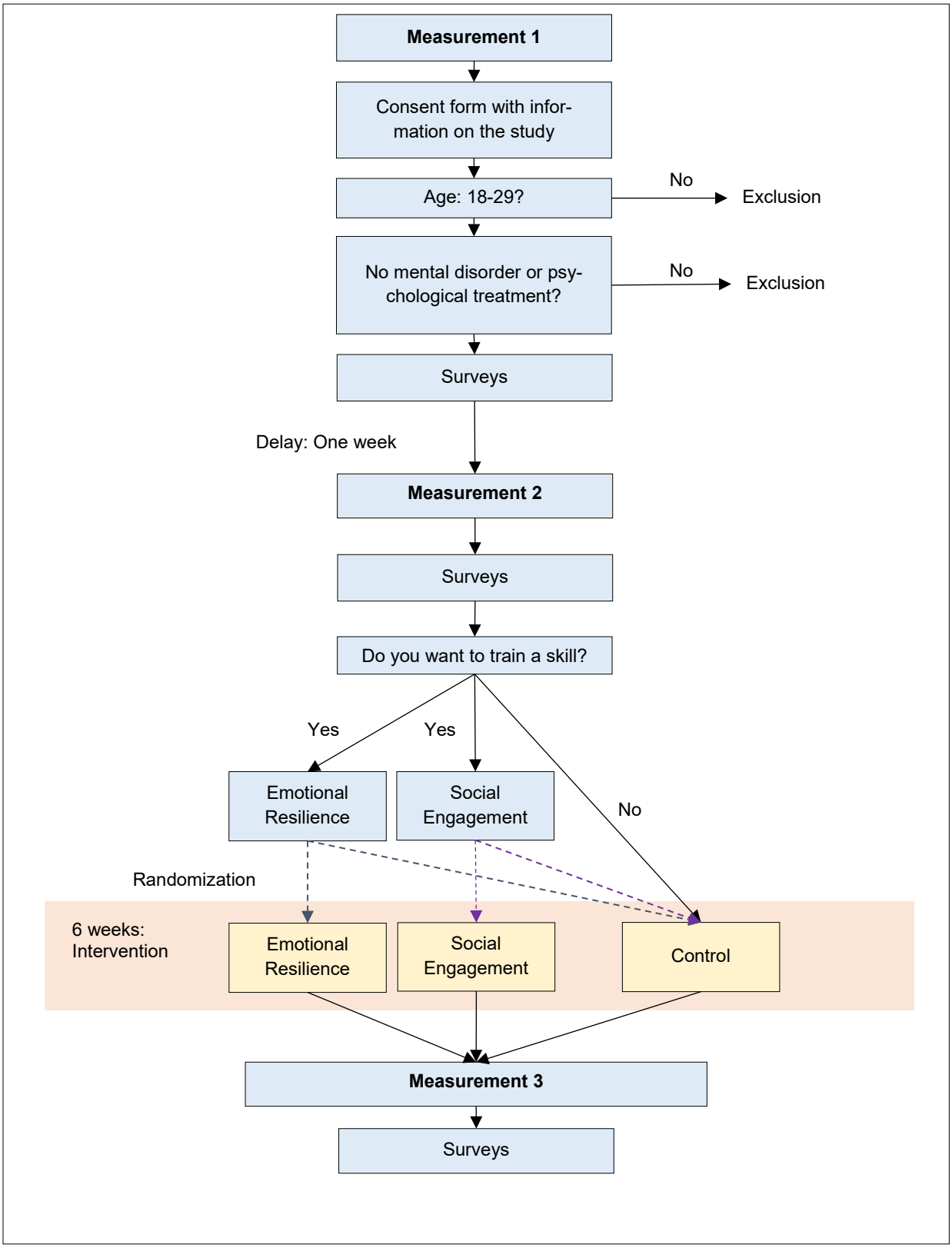
PROJECT DESIGN AND PROCEDURE

A longitudinal quasi-experimental pretest-posttest control group design was chosen with overall three measurement occasions and weekly questions for seven weeks between the second and third measurement occasions (see Figure 1).

During the second assessment, all participants receive information about SEB skills and personality traits as well as the opportunity to select a skill they would like to train by choosing between two skills (i.e., ER or SE) or opting for “no skill”. Based on their choice, they are randomly assigned to either an intervention group (IG) corresponding to their chosen skill (ER-IG, SE-IG) or to the control group (CG). Individuals without a desire to improve a skill are automatically placed in the control group (CG).

Figure 1

Study concept of the SKIPPER study project



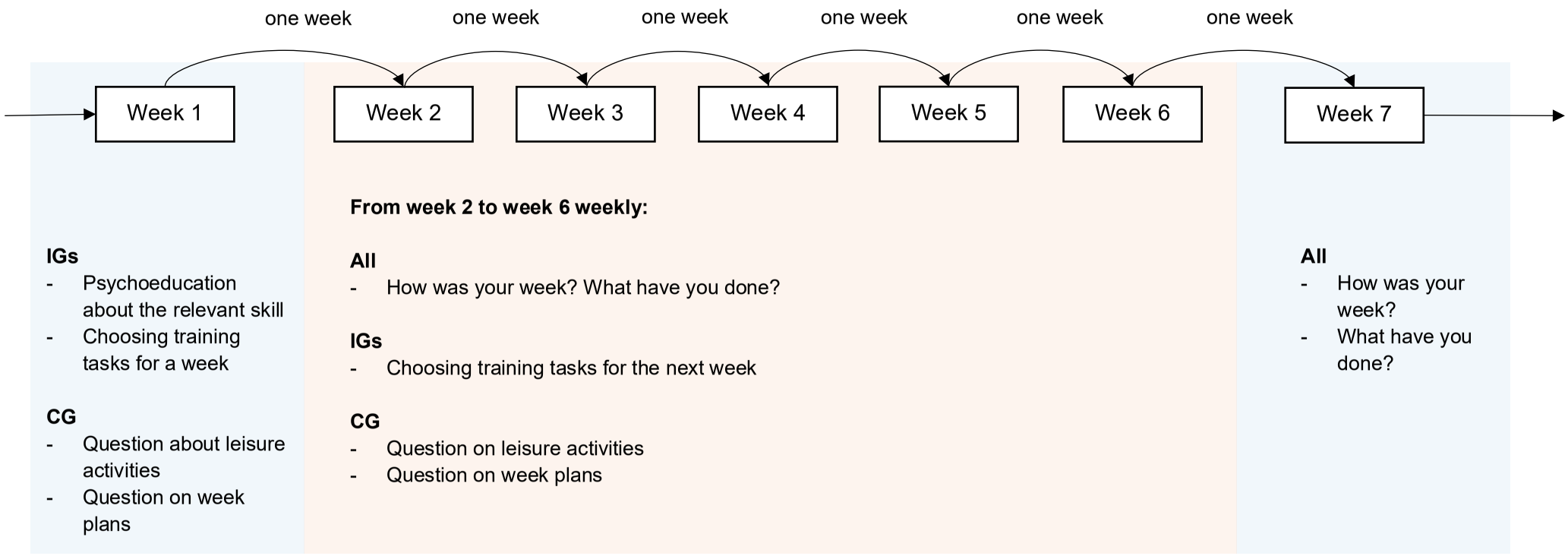
Subsequently, participants receive weekly invitations on Sundays (with reminders on Tuesdays) for six weeks to engage in skill learning – and training activities (IGs) or to provide information on their weekly plans (CG). Participants in the IGs can select predefined and explained tasks each week to practice their chosen skill, while participants in the CG are asked to share some details about their weekly plans. After each week, participants are asked to give information about their experiences of the week (see Figure 2).

Two strategies are employed to encourage participation retention and minimize dropouts (Sammut et al., 2021): At the end of the second measurement occasion, all participants can receive a feedback on their personality and skill profile, respectively. Additionally, the can receive a feedback on changes in the two skills ER and SE after the third assessment. Moreover, participants enter into a voucher lottery if they continue their participation until the end of the third measurement occasion, with the chance to win a ‘Wunschgutschein’-voucher (valued between €25 and €150).

Following the data collection, all participants are provided with information on the research objectives. If the interventions demonstrate a benefit on psychological well-being, we plan to offer them to the CG participants. The study is realized using the formR survey framework (Arslan et al., 2020) and started at 5th of November 2023. A second data collection is planned, involving a follow-up assessment to test for lasting effects.

Figure 2

Weeks between the second and third measurement occasion



INTENDED SAMPLE: TARGET GROUP

While it was recommended to examine SEB skills primarily in adolescence (Napolitano et al., 2021; Soto et al., 2022b), researchers have encouraged to investigate these skills across various developmental periods (Soto et al., 2022a). Young adulthood is momentous for development (Roberts & Davis, 2016), and yet, to the best of our knowledge, there is a lack of intervention studies addressing SEB skills within this age group. Therefore, we focus on emerging adults from Germany aged between 18 and 29 years in our research. Individuals diagnosed with a mental disorder and/or undergoing psychological treatment are excluded from participation to rule potential health risks.

MEASURES

Several psychological constructs are assessed using distinct measures. An overview of all included measures and asked questions is provided in Table 1. Additionally, some instructed items are included into the questionnaires.

Table 1

Current sequence and measures in the SKIPPER study project

Time point (T)	
T1	Welcome and consent form
T1	Demographic variables
T1	Behavioral, Emotional, and Social Skills Inventory in German language (BESSI-G, Lechner et al., 2022)
T1	Unified Motive Scales (UMS, Schönbrodt & Gerstenberg, 2012)
T1	Narcissistic Admiration and Rivalry Questionnaire (NARQ, Back et al., 2013)
T1	Big Five Inventory-2 (BFI, Danner et al., 2019)
T1	Self-esteem (3 items)
T1	Self-efficacy (Beierlein et al., 2014)
T1	Self-oriented perfectionism (Stöbner, 2012)
T1	Positive and Negative Affect Scale (PANAS, Breyer & Bluemke, 2016)
T1	Life satisfaction (L1, Beierlein et al., 2015)
T2	Questions: What is personality? What are (soft) skills? Are skills and personality the same thing in your opinion?
	BFI-2 (Danner et al., 2019)
	Psychoeducation on personality and the Big Five
T2	BESSI-G (Lechner et al., 2022)
T2	Psychoeducation on skills
T2	Quiz
T2	Question: Would you like to train one of the following skills? - Emotional Resilience Skills - Social Engagement Skills - No Question: Why (not)?
T2	Randomization based on the answer to the question about skill training
T2	Individual feedback
Six weeks	Six-week long intervention: Weekly reminder at Sundays with the opportunity to choose training exercises
T3	BESSI-G (Lechner et al., 2022)
T3	BFI-2 (Danner et al., 2019)
T3	Self-esteem (3 items)
T3	Self-efficacy (Beierlein et al., 2014)

T3	PANAS (Breyer & Bluemke, 2016)
T3	L1 (Beierlein et al., 2015)
T3	Feedback on change in Neuroticism, Extraversion, Social Engagement, and Emotional Resilience

Note. The measures could be exchanged and additional measures could be added in further research.

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